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Validity evidence for a short measure of left-wing authoritarianism among Brazilian adults and adolescents

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Abstract

Background Left-Wing Authoritarianism (LWA) has received increasing attention in recent years, yet research on the construct remains limited by the scarcity of instruments validated across different cultures and across the lifespan. Although multidimensional measures of LWA have recently been proposed, no validated Brazilian-Portuguese instrument is currently available. The present study aimed to develop and validate a short Brazilian-Portuguese measure of LWA for use with both adults and adolescents. Drawing on a multidimensional conceptualization, the proposed instrument assessed three LWA dimensions: Anti-Hierarchical Aggression (support for coercive action against dominant groups), Anti-Conventionalism (rejection of traditional norms), and Top-Down Censorship (support for authorities restricting unacceptable views).

Methods In Study 1, 764 adults completed the original 39-item LWA instrument. Exploratory Structural Equation Modeling was used to identify the three highest-loading items from each dimension, resulting in a 9-item version of the scale. In Study 2, the resulting instrument was administered to 308 adolescents, and its three-factor structure was evaluated through Confirmatory Factor Analysis.

Results Results supported the proposed three-dimensional structure across both studies, as the abbreviated measure demonstrated adequate factor loadings and internal consistency estimates. Evidence for criterion validity was also observed across both studies. All LWA dimensions were negatively associated with Right-Wing Authoritarianism and Social Dominance Orientation, and positively associated with left-wing political self-categorization. In Study 1, the LWA dimensions were also negatively associated with prejudice toward dissident and derogated groups and positively associated with pro-environmental attitudes.

Conclusions The findings thus provide evidence for the reliability and validity of a brief multidimensional measure of LWA in the Brazilian context.

Keywords Left-wing authoritarianism · Ideology · Psychometrics · Brazil

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Introduction

Evidence from more than 30 years of research describes the existence of Right-Wing Authoritarianism (RWA; Altemeyer, 1981), defined as the combination of uncritical submission to authorities perceived as legitimate, aggressiveness directed against various persons perceived as threatening or deviant, and adherence to conventional social norms. In contrast, research on Left-Wing authoritarianism (LWA) was not as effective in providing enough evidence regarding its existence (Stone & Smith, 1993), leading researchers to even consider it to be a construct "relatively unproductive and that might well be discarded" (Stone, 1980, p.14). Major obstacles to the consolidation of LWA as a robust psychological construct seem to stem from three related issues: lack of proper measurement, poor conceptualization, and asymmetric considerations in political ideology research. We begin by outlining what these three issues mean for LWA research before detailing the present study that aimed to validate a short LWA measure to the Brazilian context.

The instrumental issue

One of the major obstacles to LWA research has been the lack of psychometrically sound instruments capable of assessing it adequately. Early attempts to measure authoritarian tendencies on the political left frequently relied on adaptations of instruments originally developed to assess RWA, including the Eysenck (1954), Kohn (1972) and Wilson et al. (1976) proposed measures intended to capture authoritarian attitudes among left-wing individuals. However, as noted by Van Hiel et al. (2006), many of the instruments developed prior to the 1980s suffered from important methodological shortcomings because they were largely derived from the F-Scale proposed by Adorno et al. (1950) to measure RWA tendencies. Although historically influential, the F-Scale has been criticized on both psychometric and theoretical grounds (e.g., Duckitt et al., 2010; Funke, 2005; Mavor et al., 2010; Vilanova et al., 2020). From a psychometric perspective, concerns have been raised regarding its susceptibility to acquiescence bias, item wording, and limited evidence for the adequacy of its measurement properties. Theoretically, several items were criticized for their ambiguity and lack of specificity, making their interpretation difficult (e.g., "Familiarity breeds contempt"; Adorno et al., 1950, p. 256). A substantial advance occurred with the development of Altemeyer's (1981) Right-Wing Authoritarianism scale, which addressed many of these limitations through improved item construction and a more clearly articulated theoretical framework. Altemeyer's RWA scale subsequently became the principal instrument for the study of authoritarianism and served as the basis for later efforts to develop psychometrically stronger measures of LWA.

Building upon his work on RWA, Altemeyer (1996) later proposed one of the first explicit attempts to measure LWA. According to Altemeyer, LWA would be characterized by the covariation of three attitudinal tendencies. The first, "Authoritarian Submission", refers to a strong inclination to submit to authorities seeking to overthrow established and legitimate authorities in society. The second, "Authoritarian Aggression", reflects hostility and aggressiveness directed toward established authorities and those perceived as supporting them. Finally, "Conventionalism" involves a high degree of adherence to norms and values endorsed by revolutionary authorities. The resulting LWA scale demonstrated good internal consistency (Cronbach's $\alpha = 0.86$) and represented a substantial methodological advancement over previous measures by providing a theoretically coherent instrument with satisfactory psychometric properties. However, the Altemeyer (1996) study also gave rise to a new controversy. Among the 1,188 participants who completed the instrument, only 32 scored above the theoretical midpoint of the scale (i.e., above 5 on a response range from 1 to 9). Moreover, even among these participants, the highest observed score was only 5.85. Based on these findings, Altemeyer concluded that genuinely left-wing authoritarian individuals were exceedingly rare and suggested that LWA was, at best, an uncommon phenomenon. This observation shifted the debate surrounding LWA from the question of whether it could be measured to whether it existed in sufficient prevalence to constitute a meaningful psychological and political construct. Because this

argument relied on the proportion of individuals scoring above the theoretical midpoint of the scale, we now refer to it as the "theoretical midpoint issue", which is discussed in detail below.

The theoretical midpoint issue

Despite its satisfactory psychometric properties, Altemeyer's (1996) LWA scale was subsequently criticized on theoretical grounds. One criticism concerned the conceptual distinctiveness of the dimensions comprising the instrument. Specifically, Van Hiel et al. (2006) argued that the Conventionalism dimension substantially overlapped with Authoritarian Submission, given that adherence to norms and values endorsed by revolutionary authorities inherently entails a willingness to defer to those same authorities. From this perspective, Conventionalism may represent a consequence or manifestation of authoritarian submission rather than an independent component of LWA. A second criticism was that the defining feature of LWA lies in the simultaneous endorsement of Authoritarian Aggression and Authoritarian Submission. According to this argument, it is precisely the combination of these two tendencies that distinguishes left-wing authoritarians from other individuals occupying the political left. For example, anarchists may exhibit high levels of authoritarian aggression insofar as they strongly oppose established authorities and existing power structures. However, unlike left-wing authoritarians, anarchists are also resistant to authorities seeking to replace the existing order and therefore would not display elevated levels of authoritarian submission. Consequently, Van Hiel et al. (2006) suggested that the joint presence of aggression directed toward established authorities and submission to revolutionary authorities constitutes the core of LWA, whereas Conventionalism may play a more peripheral role in differentiating left-wing authoritarians from other left-wing ideological groups.

Van Hiel et al. (2006) then developed a revised LWA measure emphasizing what they considered to be its defining features: Authoritarian Aggression and Authoritarian Submission. To construct the new instrument, the authors retained items from Altemeyer's original scale that explicitly referred to the endorsement of violence against established authorities and obedience to revolutionary leaders, while modifying the grammatical objects of several remaining items to improve their applicability. The resulting instrument consisted of eight items and demonstrated satisfactory psychometric properties (Van Hiel et al., 2006).

Notably, Van Hiel et al. (2006) also revisited the question raised by Altemeyer's findings regarding the apparent rarity of left-wing authoritarians, or the "theoretical midpoint issue". Specifically, they hypothesized that the low prevalence observed by Altemeyer might have reflected the composition of his sample rather than the actual absence of the phenomenon, given that individuals actively involved in radical left-wing movements had not been adequately represented. To test this possibility, Van Hiel et al. (2006) assessed LWA not only among ordinary voters but also among political activists affiliated with various social movements, including Stalinist and neo-Marxist organizations. In the first study, which surveyed 208 ordinary voters, only eight participants (3.85%) scored above the theoretical midpoint of the scale. In the second study, nine of the 264 ordinary voters (3.40%) exceeded the midpoint, whereas this proportion increased to 14.50% among political activists, with 10 of the 69 activists scoring above this threshold. Importantly, unlike Altemeyer's (1996) findings, some participants in the activist sample obtained the highest possible score on the instrument.

Based on these results, Van Hiel et al. (2006) concluded that LWA should not be regarded as a non-existent construct. Rather, it appeared to be relatively uncommon among ordinary voters but more prevalent within segments of the far left engaged in political activism. Consequently, the debate surrounding LWA shifted once again. If LWA could indeed be identified among certain populations, the central question was no longer whether the construct existed, but whether its prevalence was sufficiently comparable to that of RWA to justify claims of symmetry between left- and right-wing forms of authoritarianism. This concern subsequently became known as the symmetry issue in the literature on authoritarianism.

The symmetry issue

Early research on political ideology often suggested that conservatives exhibited higher levels of prejudice, intolerance, and opposition to egalitarian values than liberals. For example, conservatism has been associated with more negative attitudes toward racial minorities (e.g., Sears & Henry, 2003), greater prejudice against homosexual individuals (e.g., Terrizzi et al., 2010), and lower endorsement of egalitarian principles (e.g., Ho et al., 2015). More recent evidence, however, has challenged the notion that judgmental biases are exclusive to one side of the political spectrum. Studies have shown that liberals and conservatives tend to exhibit biases of similar magnitude, although these biases are directed toward different targets (Chambers et al., 2012; Ditto et al., 2018). For instance, Chambers et al. (2012) found that liberals expressed more negative attitudes toward an African American target when that individual was described as politically conservative, whereas conservatives expressed more negative attitudes toward a European American target when that individual was described as politically liberal. Likewise, Ditto et al. (2018) demonstrated that motivated reasoning and partisan bias occur to comparable degrees among liberals and conservatives. Taken together, these findings suggest that certain psychological tendencies may be symmetrical across the political spectrum, differing primarily in the objects toward which they are directed. From this perspective, authoritarianism may also represent a potentially symmetrical construct, with left- and right-wing authoritarians exhibiting similar underlying tendencies that become expressed toward ideologically distinct targets and authorities.

Based on the idea that being authoritarian is not an exclusive susceptibility of conservatives, a new LWA scale has been proposed (Conway et al., 2018). Authors sought to create items that kept the semantic structure of the RWA scale (Altemeyer, 1998) but directed towards groups that are targets of LWA. Therefore, items were composed of the same expressions used in the RWA scale (e.g., "smash", "strongly punish"; Altemeyer, 1998) but directed against other groups. For the development of this new LWA scale, the items of the RWA scale (Altemeyer, 1998) were adapted to refer to uncritical submission to left-wing leaders and groups that are targets of LWA in the American context. For instance, the original RWA item "It's always better to trust the judgment of the proper authorities in government and religion than to listen to the noisy rabble-rousers in our society who are trying to create doubts in people's minds" (Altemeyer, 1998) was modified in the LWA scale to "It's always better to trust the judgment of the proper authorities in science with respect to issues like global warming and evolution than to listen to the noisy rabble-rousers in our society who are trying to create doubts in people's minds" (Conway et al., 2018, p. 7).

Conway et al. (2018) reported findings supporting the validity of their LWA measure. Specifically, higher LWA scores were associated with more liberal political self-identification, greater negativity toward Christian groups, and stronger endorsement of pro-environmental beliefs. Although these findings provided important evidence for the construct validity of the instrument, an important limitation of the measure deserves consideration. Notably, Conway et al. (2018) based the development of their scale on Altemeyer's (1996) RWA scale. Despite its widespread use, the RWA scale has long been criticized for the convoluted nature of many of its items, which often combine multiple propositions within a single statement. For example, the item "It's always better to trust the judgment of the proper authorities in government and religion than to listen to the noisy rabble-rousers in our society who are trying to create doubts in people's minds" (Altemeyer, 1998, p. 86) simultaneously assesses deference to authorities, distrust of dissenters, and attitudes toward specific social institutions. Consequently, when respondents indicate partial agreement with such items, it becomes difficult to determine which component of the statement they endorse or reject. The same criticism applies to Conway et al.'s (2018) LWA measure, as illustrated by the item "It's always better to trust the judgment of the proper authorities in science with respect to issues like global warming and evolution than to listen to the noisy rabble-rousers in our society who are trying to create doubts in people's minds" (p. 7). Thus, although Conway et al.'s instrument represented an important step forward in the contemporary study of LWA, concerns regarding item complexity and interpretability remained, motivating subsequent efforts to develop measures with clearer and more narrowly defined indicators of the construct.

To address these psychometric concerns, Costello et al. (2022) developed a revised measure of LWA consisting of 39 items distributed across three theoretically distinct dimensions. The first dimension, "Anti-Hierarchical Aggression", reflects support for coercive or punitive actions directed at individuals and groups perceived as maintaining unjust social hierarchies or obstructing progressive social change (e.g., "The rich should be stripped of their belongings and status", p. 16). The second dimension, "Anti-Conventionalism", captures the rejection of traditional norms and values and the endorsement of progressive moral conventions as self-evidently correct (e.g., "Anyone who opposes gay marriage must be homophobic", p. 16) Finally, the third dimension, "Top-Down Censorship", refers to support for restricting or suppressing the expression of views regarded as harmful, offensive, or socially unacceptable through the intervention of legitimate authorities (e.g., "University authorities are right to ban hateful speech from campus", p. 16).

Initial validation studies provided evidence for the construct validity and psychometric robustness of the instrument. For instance, a three-factor Exploratory Structural Equation Model (ESEM) demonstrated adequate fit, and the three dimensions were associated with related constructs in theoretically expected ways, including RWA and Social Dominance Orientation (SDO, Pratto et al., 1994), providing further support for the construct validity of the instrument (Costello et al., 2022, Table S11). Importantly, subsequent studies have extended these findings across diverse cultural contexts. Although the Anti-Conventionalism subscale showed weaker item loadings in a New Zealand sample, this subscale showed a similar pattern of associations to the other LWA subscales and the three-factor model had good fit to data in that sample (Milfont & Osborne, 2024). A German adaptation replicated the expected three-factor structure and demonstrated measurement invariance across representative samples from Austria, Germany, and the German-speaking part of Switzerland (Krispenz et al., 2025). Likewise, a Chilean validation study supported the structural validity of the scale and showed that an abbreviated 9-item version retained adequate reliability, factorial validity, and invariance across sex among university students (Troncoso-Tejada et al., 2025). These findings suggest that the multidimensional approach not only addresses important psychometric limitations of earlier measures but also captures distinct manifestations of left-wing authoritarian tendencies across diverse sociocultural contexts.

The present study

Based on these successful previous adaptations, the present research sought to develop and provide initial validation evidence for a brief LWA measure derived from the multidimensional instrument proposed by Costello et al. (2022). Although the original 39-item measure demonstrated promising psychometric properties, its length may limit its use in large-scale surveys, studies involving multiple constructs, and research conducted with younger populations. Accordingly, our objective was to identify a reduced set of items that adequately represented the three dimensions of the construct while preserving the validity and reliability of the original instrument. To examine the applicability of the resulting short form across developmental contexts, we conducted two studies. Study 1 involved an adult sample and focused on the selection of items and the evaluation of the factor structure, reliability, and criterion validity of the abbreviated measure. Based on previous research (Costello et al., 2022; Krispenz et al., 2025; Milfont & Osborne, 2024; Troncoso-Tejada et al., 2025), we formulated five hypotheses. First, we expected the original three-factor structure of the LWA measure to be replicated in the Brazilian context. Second, we hypothesized that the LWA dimensions would be negatively associated with political self-categorization on the left-right political spectrum, such that individuals who identified more strongly with the political right would report lower levels of LWA. Third, we hypothesized that the LWA dimensions would be negatively associated with RWA, consistent with prior findings suggesting that left- and right-wing authoritarian tendencies are anchored in distinct ideological contents (Conway et al., 2018; Costello et al., 2022; Troncoso-Tejada et al., 2025). Fourth, we expected negative associations between LWA and SDO, reflecting the anti-hierarchical orientation underlying left-wing authoritarian attitudes (Costello et al., 2022; Troncoso-Tejada et al., 2025). Fifth, we hypothesized that LWA would be positively associated with pro-environmental attitudes, in line with previous

evidence linking left-wing ideological orientations to greater environmental concern (Conway et al., 2018; Milfont & Osborne, 2024). In addition, we explored the associations between LWA and the dimensions of generalized prejudice proposed by Duckitt and Sibley (2007)—namely, prejudice toward dangerous, dissident, and derogated groups. Because these relationships have not been systematically examined using the multidimensional conceptualization of LWA, no a priori hypotheses were specified for these analyses. Finally, we also examined the prevalence of LWA by estimating the proportion of participants scoring above the theoretical midpoint of each dimension. Given the debate surrounding the prevalence of LWA (e.g., Altemeyer, 1996; Van Hiel et al., 2006) and the absence of clear benchmarks in the recent literature, these analyses were likewise considered exploratory and were conducted without specific a priori hypotheses.

Study 2 extended these analyses to an adolescent sample, providing evidence regarding the suitability of the short measure for use with younger respondents. This extension is particularly important because adolescence constitutes a critical period for the development of political orientations, intergroup attitudes, and beliefs about authority (Duckitt and Sibley 2017; Vilanova et al. 2024, 2026a). Consequently, the availability of a brief and psychometrically adequate measure of LWA for this age group may enable future research to examine when left-wing authoritarian tendencies first emerge, how they evolve across development, and which individual and contextual factors contribute to their expression. Consistent with Study 1, we formulated four hypotheses for the adolescent sample. First, we hypothesized that the original three-factor structure of the LWA measure would be replicated among adolescents, providing evidence that the multidimensional conceptualization of LWA is applicable beyond adult populations. Second, we expected a negative association between LWA score and political self-categorization on the left-right political spectrum. Third, we expected the LWA dimensions to be negatively associated with RWA. Fourth, we hypothesized that negative associations between LWA and SDO would also be observed. Finally, as in Study 1, we explored the prevalence of LWA by estimating the proportion of adolescents scoring above the theoretical midpoint on each dimension. Given the limited literature examining LWA during adolescence and the absence of established prevalence estimates for this age group, no specific hypotheses were formulated regarding the magnitude of these prevalence rates.

Study 1

Method

Participants

Originally, 854 individuals responded to the survey, but 70 were excluded for being under 18 years old and 20 were excluded for completing less than 80% of the survey items (Schlomer et al., 2010). A total of 764 adult Brazilian participants were included in the final sample, and 521 (68.2%) identified as female. Ages ranged from 18 to 80 years old ($M = 35.36$; $SD = 14.80$). Participants were recruited through convenience sampling via a link advertised on Instagram inviting to participate in a survey about social attitudes in October 2021. Before answering the instruments, participants expressed their consent by providing their agreement to an Informed Consent Form. Anonymity was granted and only researchers had access to the data.

Measures

Participants completed a translated version of the LWA scale originally proposed by Costello et al. (2022). As outlined in the Introduction section, the scale comprises three dimensions: Anti-Hierarchical Aggression, Anti-Conventionalism, and Top-Down Censorship. Responses were given on a 7-point agreement scale ranging from 1 (totally disagree) to 7 (totally agree).

In addition to the LWA scale, participants completed a sociodemographic questionnaire (see Table 1) assessing socioeconomic level, schooling, ethnicity, civil status, if the person is religious or not, and political self-categorization. Participants also responded to the Authoritarianism–Conservatism–Traditionalism (ACT) scale indexing RWA, originally developed by Duckitt et al. (2010) and adapted to the Brazilian context by Vilanova et al. (2018). The scale comprises 34 items rated on a 5-point Likert scale ranging from 1 (totally disagree) to 5 (totally agree), and demonstrated good internal consistency in the present sample (Cronbach's $\alpha = 0.92$, 95% CI [0.91, 0.93]; McDonald's $\omega = 0.923$, 95% CI [0.916, 0.931]). Additionally, participants completed the 16-item SDO_7 scale (Ho et al., 2015), adapted to the Brazilian context by Vilanova et al. (2022a). Items were rated on a 7-point agreement scale (1 = totally disagree to 7 = totally agree), with good internal consistency ($\alpha = 0.882$, 95% CI [0.866, 0.897]; $\omega = 0.885$, 95% CI [0.874, 0.896]). Generalized prejudice was then assessed using the affective thermometer originally proposed by Asbrock et al. (2010) and adapted to the Brazilian context by Cantal et al. (2015); Vilanova et al. (2022b). Participants rated their feelings toward 25 groups on a 7-point scale, with higher scores indicating more negative affect; internal consistency was good ($\alpha = 0.873$, 95% CI [0.860, 0.885]; $\omega = 0.842$, 95% CI [0.829, 0.855]). Finally, participants completed the revised 15-item New Environmental Paradigm (NEP) scale (Dunlap

Table 1 Sociodemographic characterization of the sample (Study 1)

Characteristic	*n* (%)
	Total (*N* = 764)
Schooling	
Primary School	11 (1.4)
Secondary School	201 (26.3)
University/College	394 (51.6)
Post-Graduate	157 (20.5)
Do not know	1 (0.1)
Ethnicity	
White	499 (65.3)
Black	61 (8)
Pardo	189 (24.7)
Yellow/Asian	13 (1.7)
Indigenous	2 (0.3)
Socioeconomic level	
Monthly income equal to or higher than 20 Brazilian minimum wage	30 (3.9)
Monthly income from 10 up to 20 Brazilian minimum wages	77 (10.1)
Monthly income from 4 up to 10 Brazilian minimum wages	215 (28.1)
Monthly family income from 2 up to 4 Brazilian minimum wages	192 (25.1)
Monthly family income up to 2 Brazilian minimum wages	194 (25.4)
Do not know	56 (7.3)
Civil Status	
Single	442 (57.9)
Married	134 (17.5)
Cohabiting	101 (13.2)
Divorced	67 (8.8)
Widowed	10 (1.3)
Other	10 (1.3)
Religious	
Yes	480 (62.8)
No	284 (37.2)
Political self-categorization	
Left	256 (33.5)
Center-Left	143 (18.7)
Center	39 (5.1)
Center-Right	38 (5)
Right	58 (7.6)
None of the aforementioned	230 (30.1)

et al., 2000), adapted to the Brazilian context by Pires et al. (2016). Responses ranged from 1 (totally disagree) to 5 (totally agree), with higher scores reflecting stronger pro-environmental attitudes. The scale demonstrated acceptable internal consistency in this sample (α = 0.77, 95% CI [0.74, 0.80]; ω = 0.773, 95% CI [0.750, 0.796]).

Adaptation procedures

The original 39-item scale was first translated into Brazilian Portuguese by two independent political psychologists who were native speakers of Brazilian Portuguese, fluent in English, and experienced in psychological assessment and cross-cultural test adaptation. Both translators were also familiar with the literature on authoritarianism. Discrepancies between translations were resolved by a third expert in political psychology, likewise a native speaker of Brazilian Portuguese and residing in an anglophone country. The reconciled version was subsequently back-translated, and the back-translation was reviewed and approved by the scale's original first author (i.e., Costello). Throughout all stages, the guidelines for cross-cultural adaptation of psychological instruments proposed by the International Test Commission (2018) were followed, with particular attention to conceptual and idiomatic equivalence. Content validity was established through expert consensus at each stage of the adaptation process.

Data analysis

To identify the items with the strongest psychometric properties for each LWA dimension, the same methodological procedures used by the original authors (Costello et al., 2022) were used. Hence, first an ESEM using the Weighted Least Squares Mean and Variance adjusted (WLSMV) estimator with Target rotation was conducted using Mplus version 6.12. ESEM combines the exploratory nature of Exploratory Factor Analysis (EFA) with the broader modeling framework of Structural Equation Models. Unlike conventional EFA, which is primarily used to explore the underlying factor structure, ESEM allows researchers to specify an a priori factor structure while freely estimating cross-loadings through rotation, thereby providing a more flexible assessment of multidimensional constructs (Asparouhov & Muthén, 2009; Marsh et al., 2014). This approach was adopted in the present study because the original LWA measure was developed and evaluated using ESEM, and we sought to maintain consistency with the original validation approach. The three items with the highest factor loadings in each dimension were retained for the short version of the LWA scale. Specifically, three items were selected for each dimension, as a minimum of three indicators per latent factor has been recommended to ensure adequate identification and stability of factor solutions (Bollen, 1989; Raubenheimer, 2004). Model fit was evaluated using the following criteria: $\chi^2/df < 2$, RMSEA ≤ 0.06, CFI and TLI ≥ 0.95 (Bollen & Long, 1993; Hu & Bentler, 1999). After the three top-loading items were identified through ESEM, a Confirmatory Factor Analysis (CFA) using the WLSMV estimation method was conducted to assess its model fit. To assess criterion validity, Pearson correlations were calculated between the mean scores of each LWA dimension and political self-categorization. Additionally, associations between the LWA dimensions and measures of RWA, SDO, generalized prejudice dimensions, and pro-environmental attitudes were examined. Then, the reliability of each dimension was assessed using Cronbach's α and McDonald's ω. Finally, the prevalence of LWA was assessed by estimating the proportion of participants scoring above the theoretical midpoint of each dimension (i.e., 4).

Results and discussion

Cross-cultural adaptation

Disagreements between the independent translators emerged in 8 of the 39 items, primarily reflecting different translation strategies: one prioritized close fidelity to the original wording, whereas the other proposed adaptations to capture broader manifestations of LWA in the Brazilian context. All discrepancies were resolved by the

third expert. Across all cases, the selected versions favored the first strategy, i.e., closer semantic equivalence to the original items while preserving conceptual clarity in the target language. Specifically, retained translations maintained the original referents, targets, and behavioral scope, avoiding broader or reinterpretive reformulations that altered item meaning. Full details of the discrepancies and final decisions are presented in Table 2.

Validity evidence of the LWA scale

Table 3 presents the rotated factor loadings obtained through ESEM for all items. The three-factor ESEM model showed a mixed pattern of fit indices, with RMSEA meeting the predefined criterion for adequate fit, although χ^2/df, CFI, and TLI fell short of the more stringent thresholds specified a priori ($\chi^2 = 2006.847$, $df = 627$; $\chi^2/df = 3.20$; CFI = 0.943; TLI = 0.932; RMSEA = 0.054, 90% C.I. [0.51, 0.56], WRMR = 1.212). Hence, for Anti-Hierarchical Aggression, the items with the highest factor loadings were respectively "*as pessoas ricas deveriam ser forçadas a renunciar a praticamente toda a sua fortuna* (rich people should be forced to give up virtually all of their wealth)" ($\lambda = 0.977$), "*os ricos deveriam ter seus pertences e status social removidos* (the rich should be stripped of their belongings and status)" ($\lambda = 0.935$), and "*se eu pudesse refazer a sociedade*, *colocaria as pessoas que atualmente têm mais privilégio nas posições sociais mais baixas* (if I could remake society, I would put people who currently have the most privilege at the very bottom)" ($\lambda = 0.746$). For Anti-Conventionalism, the top-loading items were "*no fundo*, *praticamente todas as pessoas politicamente conservadoras são racistas*, *sexistas e homofóbicas* (deep down, just about all conservatives are racist, sexist, and homophobic)" ($\lambda = 0.796$), "*pessoas que são politicamente conservadoras são moralmente inferiores* (conservatives are morally inferior)"

Table 2 Translation discrepancies and final decisions for the LWA scale (Study 1)

Original item	Translation A	Translation B	Retained version
Schools should be required by law to teach children about our country's history of racism, classism, sexism, and homophobia	As escolas deveriam ser obrigadas por lei a ensinarem às crianças a história de racismo, classismo, sexismo e homofobia em nosso país	Deveria haver uma lei para banir das escolas autores clássicos com posições que hoje são tidas como ofensivas	A
Anyone who opposes gay marriage must be homophobic	Qualquer pessoa que se opõe ao casamento gay deve ser homofóbica	Todas as pessoas religiosas são preconceituosas	A
People who are truly worried about terrorism should shift their focus to the nutjobs on the far-right	Pessoas realmente preocupadas com terrorismo deveriam mudar o foco para os lunáticos da extrema-direita	Atos públicos de grupos de direita devem ser encarados como terrorismo	A
University authorities are right to ban hateful speech from campus	As autoridades universitárias estão certas em banir discursos de ódio do campus	As autoridades universitárias estão certas em banir do campus atividades que sejam ofensivas	A
I am in favor of allowing the government to shut down right-wing internet sites and blogs that promote nutty, hateful positions	Sou a favor de permitir que o governo possa desativar sites e blogs mantidos por pessoas de direita que promovem opiniões fora da realidade e de ódio	Sou a favor de permitir que o governo possa desativar sites e blogs mantidos por grupos de direita	A
Even books that contain racism or racial language should not be censored	Até mesmo livros que contêm racismo ou linguagem racial não deveriam ser censurados	Livros clássicos que contêm visões hoje tidas como ofensivas sobre raça não deveriam ser censurados	A
I don't support shutting down speakers with sexist, homophobic, or racist views	Não apoio o cancelamento de palestrantes com visões sexistas, homofóbicas ou racistas	Não apoio o cancelamento de palestrantes cujas opiniões sobre gênero, sexualidade e raça sejam ofensivas	A
Neo-Nazis ought to have a legal right to their opinions	Neonazistas devem ter o direito legal de manifestar suas opiniões	Grupos de extrema-direita devem ter o direito legal de manifestar suas opiniões	A

Table 3 Rotated factor loadings obtained through ESEM (Study 1)

Item	AHA	AC	TDC
Rich people should be forced to give up virtually all of their wealth	**0.977**	−0.214	0.161
The rich should be stripped of their belongings and status	**0.935**	−0.203	0.155
If I could remake society, I would put people who currently have the most privilege at the very bottom	**0.746**	−0.063	0.026
Our country would be much better off if all wealthy people were at the bottom of the social hierarchy	0.662	−0.132	0.133
When the tables turn on the oppressors at the top of society, I will take pleasure in seeing them suffer the violence they inflicted on so many others	0.594	0.136	−0.110
Most wealthy executives deserve to be thrown in prison	0.561	0.181	−0.074
Constitutions and laws are just another way for the powerful to destroy our dignity and individuality	0.453	0.122	−0.198
The current socioeconomic system is beyond repair	0.279	0.234	−0.100
We need to replace the established order at any cost	0.504	0.157	−0.020
Political violence can be constructive when it serves the cause of social justice	0.525	0.111	−0.109
Certain people in our society must pay for the violence committed by their ancestors	0.434	0.278	−0.031
If some of the worst right-wing politicians were assassinated, it wouldn't be the end of the world	0.416	0.352	−0.074
I would prefer a far-left leader with absolute authority to a right-wing leader with limited powers	0.439	0.173	0.085
Deep down, just about all conservatives are racist, sexist, and homophobic	−0.001	**0.796**	0.065
Conservatives are morally inferior	0.099	**0.765**	−0.007
I hate being around conservative people	0.007	**0.762**	0.117
Schools should be required by law to teach children about the history of racism, classism, sexism, and homophobia in our country	0.064	0.502	0.331
Anyone who opposes gay marriage must be homophobic	0.013	0.616	0.172
I can't imagine myself becoming friends with someone who is politically conservative	0.035	0.745	0.119
People genuinely concerned about terrorism should shift their focus to far-right lunatics	0.179	0.555	0.131
Traditional ways of life and values need to be abolished	0.206	0.552	0.087
Radical and progressive moral values can save our society	0.313	0.148	0.063
All politically conservative people are idiots	0.108	0.734	0.022
It is important to destroy the nationalist and imperialist values of Western countries	0.286	0.488	0.047
I try to expose myself to politically conservative news sources	0.262	−0.098	−0.213
There is nothing wrong with biblical retreats	−0.022	−0.435	0.105
I should have the right not to be exposed to offensive views	−0.010	−0.234	**0.853**
University authorities are right to ban hateful speech from campus	−0.118	0.095	**0.809**
To succeed, a workplace must ensure that its employees feel safe from criticism	0.022	−0.140	**0.637**
Classrooms should be safe spaces for discussion that protect students from offensive opinions	−0.138	0.101	0.501
We must obey strong leaders who have the will to eliminate prejudice and intolerance	0.090	−0.017	0.480
Defending the right to freedom of expression protects the rights of sexists, racists, and homophobes at the expense of marginalized people	0.095	0.186	0.236
I am in favor of allowing the government to shut down websites and blogs maintained by right-wing individuals that promote hateful views	0.001	0.282	0.584
Universities that allow speakers with offensive views should be publicly condemned	0.199	0.074	0.574
Eliminating inequality is more important than protecting the so-called right to freedom of expression	0.095	0.201	0.388
Right-wing radio stations and other politically conservative media outlets should be prohibited from broadcasting their opinions	0.209	0.452	0.221
Even books containing racism or racial language should not be censored	−0.063	0.132	−0.317
I do not support the cancellation of speakers with sexist, homophobic, or racist views	−0.037	−0.115	−0.307
Neo-Nazis should have the legal right to express their opinions	0.119	−0.263	−0.454

Top-three factor loadings per dimension are in bold. Item wording may differ slightly from the original English version because the items are presented as back-translations of the Portuguese adaptation rather than as verbatim reproductions of the original instrument

AHA Anti-Hierarchical Aggression, *AC* Anti-Conventionalism, *TDC* Top-Down Censorship

(λ=0.765), and "*odeio estar entre pessoas politicamente conservadoras* (I hate being around conservative people)" (λ=0.762). Finally, for Top-Down Censorship top-loading items were "*eu deveria ter o direito de não ser exposto a opiniões ofensivas* (I should have the right not to be exposed to offensive views)" (λ=0.853), "*as autoridades universitárias estão certas em banir discursos de ódio do campus* (university authorities are right to ban hateful speech from campus) (λ=.809), and "*para ter sucesso, um local de trabalho deve garantir que*

seus funcionários se sintam protegidos de críticas (to succeed, a workplace must ensure that its employees feel safe from criticism") (λ = .637). Latent inter-factor correlations for the three-item scores were moderate, with Anti-Hierarchical Aggression and Anti-Conventionalism showing the strongest association (Φ = .65), followed by the association between Anti-Conventionalism and Top-Down Censorship (Φ = .42) and then Anti-Hierarchical Aggression and Top-Down Censorship (Φ = .31). CFA indicated mixed fit for this structure, as CFI was adequate but other fit indices were inadequate considering the thresholds specified a priori (χ^2 = 108.468, *df* = 24; χ^2/df = 4.52; CFI = .950; TLI = .921; RMSEA = .067, 90% C.I. [.55, .81]).

Reliability of the short LWA scale for adults

Cronbach's α and McDonald's ω values were adequate for Anti-Hierarchical Aggression (α = 0.80, 95% CI [0.77, 0.83]; ω = 0.82, 95% CI [0.80, 0.84]) and Anti-Conventionalism (α = 0.79, 95% CI [0.77, 0.82]; ω = 0.80, 95% CI [0.77, 0.82]). The Top-Down Censorship dimension presented lower but still acceptable reliability indices (α = 0.68, 95% CI [0.64, 0.72]; ω = 0.73, 95% CI [0.70, 0.77]), which may reflect the more heterogeneous content of the items comprising this dimension.

Criterion validity of the short LWA scale for adults

Distinct patterns of associations provided evidence for the criterion validity of the 9-item LWA measure. First, Pearson correlations indicated that more left-wing political self-categorization was associated with higher scores across all three LWA dimensions. For this specific analysis, participants who selected "none of the aforementioned" in the political self-categorization item (*n* = 230) were excluded, resulting in a sample of *n* = 534. Consistent with expectations, right-wing self-categorization was negatively correlated with Anti-Conventionalism (*r* (532) = −0.61, *p* < .001, 95% CI [−0.66, −0.56]), Anti-Hierarchical Aggression (*r* (532) = −0.44, *p* < .001, 95% CI [−0.51, −0.37]), and Top-Down Censorship (*r* (532) = −0.27, *p* < .001, 95% CI [−0.35, −0.19]). These findings support the criterion validity of the short LWA scale, as all three dimensions were associated with political self-categorization in the expected direction. Second, as shown in Table 4, all three LWA dimensions were negatively associated with RWA, SDO, prejudice toward derogated groups, and prejudice toward dissident groups, while being positively associated with pro-environmental attitudes, providing support for our hypotheses. One noteworthy exception was the Top-Down Censorship dimension, which exhibited a positive association with prejudice toward dangerous groups (*r* (762) = 0.12, *p* = .001, 95% CI [0.05, 0.19]). Although this finding was not

Table 4 Associations between dimensions of left-wing authoritarianism, right-wing authoritarianism, social dominance orientation, negative attitudes toward dangerous, derogated, and dissident groups, and pro-environmental attitudes (Study 1)

Variable	AHA	AC	TDC	RWA	SDO	Prejudice (Dangerous)	Prejudice (Derogated)	Prejudice (Dissident)
AHA	—							
AC	0.51**	—						
TDC	0.16**	0.27**	—					
RWA	−0.42**	−0.56**	−0.11*	—				
SDO	−0.28**	−0.34**	−0.31**	0.52**	—			
Prejudice (Dangerous)	−0.20**	−0.10*	0.12*	0.33**	0.05	—		
Prejudice (Derogated)	−0.10*	−0.06	−0.12**	0.12*	0.30**	0.04	—	
Prejudice (Dissident)	−0.31**	−0.44**	−0.17**	0.65**	0.51**	0.22**	0.50**	—
Pro-environmentalism (NEP)	0.18**	0.31**	0.22**	−0.39**	−0.45**	0.02	−0.21**	−0.41**

AHA Anti-Hierarchical Aggression, *AC* Anti-Conventionalism, *TDC* Top-Down Censorship, *RWA* Right-Wing Authoritarianism, *SDO* Social Dominance Orientation, *NEP* New Environmental Paradigm

*p < .05. **p < .001

anticipated, it may suggest that support for restricting the expression of certain groups perceived as threatening or harmful is partly linked to negative attitudes toward those groups.

Prevalence of left-wing authoritarianism among adults

Following previous studies that estimated the prevalence of authoritarian dispositions based on the proportion of individuals scoring above the theoretical midpoint of the scale (e.g., Altemeyer, 1996; Van Hiel et al., 2006), we examined the prevalence of each LWA dimension in the adult sample. Results indicated that 18.3% of participants ($n = 140$) scored above the midpoint on Anti-Hierarchical Aggression, 36.4% ($n = 278$) scored above the midpoint on Anti-Conventionalism, and 75.9% ($n = 580$) scored above the midpoint on Top-Down Censorship. For comparative purposes, the same procedure was applied to RWA and SDO scores, revealing that 14.7% of participants ($n = 112$) scored above the midpoint on mean RWA, whereas only 4.3% ($n = 32$) scored above the midpoint on mean SDO. Thus, although endorsement above the midpoint was relatively uncommon for Anti-Hierarchical Aggression and RWA, and particularly rare for SDO, support for Top-Down Censorship was observed among a substantial majority of participants.

Study 2

Method

Participants

A convenience sample was recruited as part of a broader research project investigating social attitudes in Brazil. Participants were recruited through advertisements disseminated on Instagram in October 2025, which invited adolescents to participate in an online survey about the social attitudes of Brazilian youth. Initially, 761 individuals responded to the survey. Of these, 446 were excluded for completing less than 80% of the questionnaire (Schlomer et al., 2010), and 7 were excluded for providing nonsensical responses in the age field, resulting in a final sample of 308 participants. The sample comprised 48.4% female participants aged 11–19 years ($M = 17.09$, $SD = 1.46$). Detailed sociodemographic characteristics, including educational level, ethnicity, and socioeconomic status, are reported in Vilanova et al. (2026b, Table S3). Among the retained participants, 23 had completed between 80% and 99% of the questionnaire; missing responses for these participants were imputed using the series mean method. All data collection procedures, including the waiver of written informed consent from participants' legal guardians, were conducted in accordance with the procedures approved by the Institutional Review Board under registration number CAAE [BLINDED FOR PEER REVIEW] and in compliance with Resolution No. 510/2016 of the Brazilian National Health Council. Adolescents provided informed consent before completing the survey, and access to the data was restricted to the research team.

Measures

Because the present study was embedded within a broader project on social attitudes, only the instruments relevant to the hypotheses under investigation are described below. First, participants answered a sociodemographic questionnaire. Then, participants answered the 12-item ACT scale indexing RWA, originally proposed by Duckitt et al. (2010) and adapted to the Brazilian context by Vilanova et al. (2023), with evidence of validity for the adolescent population (Vilanova et al. 2026c). Responses ranged on a scale from 1 (*strongly disagree*) to 5 (*strongly agree*). The instrument showed a good index of internal consistency in the present sample ($\alpha = 0.853$, 95% CI [0.828, 0.878]; $\omega = 0.848$, 95% CI [0.825, 0.871]). Afterward, participants completed the 8-item SDO_7 scale (Ho et al., 2015), adapted to the Brazilian context by Vilanova et al. (2022a). Items were rated on a 7-point agreement

scale (1 = totally disagree to 7 = totally agree), with good internal consistency (α = 0.918, 95% CI [0.899, 0.936]; ω = 0.919, 95% CI [0.906, 0.933]). Finally, participants completed the 9-item Left-Wing Authoritarianism Scale provided in Study 1. Responses were given on a 7-point agreement scale ranging from 1 (totally disagree) to 7 (totally agree). Reliability indices for each dimension are provided below.

Adaptation procedures

Before administering the questionnaire to the adolescent sample, a pilot study was conducted with three male adolescents aged 13, 15, and 17 years, all enrolled in public schools and with no history of grade retention. Participants completed the 9-item version of the instrument used in Study 1 and were asked to indicate whether the items were clear, explain in their own words what each item meant, and suggest possible modifications to the wording if necessary. All participants reported that the items were clear, provided interpretations consistent with the intended meaning of the items, and did not suggest any modifications. Accordingly, the same 9-item version developed in Study 1 was retained for use in the adolescent sample.

Data analysis

First, a Confirmatory Factor Analysis (CFA) was conducted using the WLSMV estimation method in RStudio version 2023.06.1. As in Study 1, model fit was evaluated according to the following criteria: $\chi^2/df < 2$, RMSEA ≤ 0.06, and CFI and TLI ≥ 0.95 (Bollen & Long, 1993; Hu & Bentler, 1999). Next, criterion validity was assessed by calculating Pearson correlations between the mean scores of each LWA dimension and political self-categorization. Associations between the LWA dimensions, RWA, and SDO were also examined. The reliability of each dimension was then evaluated using Cronbach's α and McDonald's ω coefficients. Finally, prevalence estimates for the LWA dimensions were calculated using the theoretical midpoint of the scales as a reference.

Results and discussion

Validity evidence of the short LWA scale for youth

Similarly to the results observed in the adult sample, CFA indicated that the three-dimensional structure of the 9-item LWA was adequate for the adolescent sample. Notably, the fit indices were good ($\chi^2 = 21.398$, $df = 24$; $\chi^2/df = 0.89$; CFI = 1; TLI = 1; RMSEA = 0.001, 90% C.I. [0.000, 0.041], SRMR = 0.039) and factor loadings ranged from 0.61 to 0.93 (see Table 5). Latent inter-factor correlations were moderate to strong, with Anti-Hierarchical Aggression and Anti-Conventionalism showing the strongest association (Φ = 0.72), followed by the association between Anti-Conventionalism and Top-Down Censorship (Φ = 0.71), and then Anti-Hierarchical Aggression and Top-Down Censorship (Φ = 0.58). As requested by the reviewers, discriminant validity was further examined using the Fornell–Larcker criterion (Fornell & Larcker, 1981), which evaluates whether a construct shares more

Table 5 Factor loadings obtained through confirmatory factor analysis (Study 2)

Item	AHA	AC	TDC
Rich people should be forced to give up virtually all of their wealth	0.938	-	-
The rich should be stripped of their belongings and status	0.902	-	-
If I could remake society, I would put people who currently have the most privilege at the very bottom	0.614	-	-
Conservatives are morally inferior	-	0.823	-
Deep down, just about all conservatives are racist, sexist, and homophobic	-	0.881	-
I hate being around conservative people	-	0.886	-
I should have the right not to be exposed to offensive views	-	-	0.798
University authorities are right to ban hateful speech from campus	-	-	0.814
To succeed, a workplace must ensure that its employees feel safe from criticism	-	-	0.631

AHA Anti-Hierarchical Aggression, *AC* Anti-Conventionalism, *TDC* Top-Down Censorship

Table 6 Associations between dimensions of left-wing authoritarianism, right-wing authoritarianism, and social dominance orientation (Study 2)

Variable	AHA	AC	TDC	RWA
AHA	—			
AC	0.65**	—		
TDC	0.49**	0.60**	—	
RWA	−0.57**	−0.66**	−0.44**	—
SDO	−0.57**	−0.66**	−0.66**	0.67**

AHA Anti-Hierarchical Aggression, *AC* Anti-Conventionalism, *TDC* Top-Down Censorship, *RWA* Right-Wing Authoritarianism, *SDO* Social Dominance Orientation

**$p < .001$

variance with its own indicators than it shares with other constructs. Under this criterion, the average variance extracted (AVE) for each factor should exceed the squared correlation between that factor and each other factor. AVE was 0.690 for Anti-Hierarchical Aggression, 0.746 for Anti-Conventionalism, and 0.566 for Top-Down Censorship. Each factor's AVE exceeded its highest squared latent correlation with another factor (0.518 for Anti-Hierarchical Aggression, 0.518 for Anti-Conventionalism, and 0.504 for Top-Down Censorship). Thus, all three factors met the Fornell–Larcker criterion, providing additional evidence of discriminant validity in the adolescent sample.

Reliability of the short LWA scale for youth

Cronbach's α and McDonald's ω coefficients were slightly higher than those observed in the adult sample, indicating adequate internal consistency across all dimensions. Reliability indices were satisfactory for Anti-Hierarchical Aggression (α = 0.84, 95% CI [0.80, 0.88]; ω = 0.87, 95% CI [0.85, 0.90]), Anti-Conventionalism (α = 0.90, 95% CI [0.88, 0.92]; ω = 0.90, 95% CI [0.88, 0.92]), and Top-Down Censorship (α = 0.78, 95% CI [0.73, 0.84]; ω = 0.79, 95% CI [0.74, 0.83]).

Criterion validity of the short LWA scale for youth

Similarly to Study 1, Pearson correlations indicated that more left-wing political self-categorization was associated with higher scores across all three LWA dimensions. For this analysis, participants who selected "none of the aforementioned" in the political self-categorization item ($n = 27$) were excluded, resulting in a sample of $n = 281$. Consistent with expectations, right-wing self-categorization was negatively correlated with Anti-Hierarchical Aggression (r (279) = −0.66, $p < .001$, 95% CI [−0.71, −0.59]), Anti-Conventionalism (r (279) = −0.77, $p < .001$, 95% CI [-0.82, −0.71]), and Top-Down Censorship (r (279) = −0.66, $p < .001$, 95% CI [−0.75, −0.57]). These findings support the criterion validity of the short LWA scale, as all three dimensions were associated with political self-categorization in the expected direction. Second, as shown in Table 6, all LWA dimensions were negatively associated with RWA and SDO, replicating the patterns observed in Study 1.

Prevalence of left-wing authoritarianism among youth

Based on the proportion of individuals scoring above the theoretical midpoint of the scale, 37.0% of adolescents ($n = 114$) scored above the midpoint on Anti-Hierarchical Aggression, 62.7% ($n = 193$) scored above the midpoint on Anti-Conventionalism, and 77.3% ($n = 238$) scored above the midpoint on Top-Down Censorship. Comparatively, results indicated that 17.9% of participants ($n = 55$) scored above the midpoint on mean RWA, whereas 14.6% ($n = 45$) scored above the midpoint on mean SDO. Thus, although endorsement above the midpoint was relatively uncommon for RWA and SDO, substantial proportions of adolescents endorsed Anti-Hierarchical

Aggression and Anti-Conventionalism, with Top-Down Censorship emerging as the most prevalent dimension of LWA in this sample.

General discussion

The present findings provide evidence for the validity and reliability of a brief measure of left-wing authoritarianism derived from the instrument proposed by Costello et al. (2022) in both adult (Study 1) and adolescent (Study 2) samples. Across both studies, the original three-factor structure comprising Anti-Hierarchical Aggression, Anti-Conventionalism, and Top-Down Censorship was successfully replicated, with satisfactory factor loadings and internal consistency estimates. Furthermore, the short form exhibited theoretically meaningful patterns of associations with external variables, supporting its criterion validity. Specifically, the dimensions of LWA showed associations with political self-categorization, RWA, SDO, generalized prejudice, and pro-environmental attitudes that were consistent with those reported in previous investigations using the original instrument and related measures of LWA. Importantly, these findings extended beyond adult populations, as similar validity evidence and psychometric properties were observed among adolescents. Taken together, the results suggest that the abbreviated 9-item measure retains adequate conceptual breadth and psychometric adequacy of the original scale while offering a substantially more economical alternative for use in contexts in which questionnaire length is a concern, such as large-scale surveys, studies including multiple constructs, and research involving younger respondents.

Considering the criteria commonly invoked in debates regarding the existence and relevance of LWA (e.g., Altemeyer, 1996; Conway et al., 2018; Costello et al., 2022; Van Hiel et al., 2006), the present findings provide support for its construct validity. First, the hypothesized three-dimensional structure exhibited theoretically meaningful associations with established constructs, including RWA, SDO, prejudice, environmental attitudes, and political self-categorization. These findings address one of the major criticisms historically directed at the study of LWA—namely, the absence of psychometrically sound instruments capable of assessing the construct in a multidimensional manner. Second, the results contribute to ongoing discussions regarding the prevalence of LWA. Using the theoretical midpoint as a reference, at least one of the three dimensions showed prevalence rates exceeding 18% across both studies. These rates were comparable to, and in some instances exceeded, those observed for RWA and SDO in the same samples, as well as those reported in previous studies using alternative measures of LWA (e.g., Altemeyer, 1996; Van Hiel et al., 2006). Taken together, these findings suggest that LWA is not merely a theoretical possibility or a statistically negligible phenomenon, but rather a measurable set of attitudes that can be reliably identified across developmental stages and that exhibits meaningful relations with a broader nomological network of political attitudes and beliefs.

Despite the findings supporting the validity and non-negligible prevalence of LWA, these results should be interpreted with caution. In particular, the prevalence of a psychological disposition should not be conflated with its expression as an organized political force. Although some dimensions of LWA were more prevalent than RWA in the present studies, there is limited evidence that major contemporary political parties advocate policies closely resembling the most extreme contents of the Anti-Hierarchical Aggression dimension, such as the expropriation of wealth or the removal of social status from rich people (Saunders & Jost, 2023). By contrast, analyses of party programs suggest that proposals emphasizing harsher punitive measures, stronger law-and-order policies, and the expansion of coercive state authority have historically been more characteristic of mainstream right-wing parties (Tarouco & Madeira, 2013). Consequently, the prevalence of authoritarian attitudes at the individual level does not necessarily translate into equivalent political influence, institutional power, or societal impact (Vilanova & Azevedo, 2026). Distinguishing between the distribution of psychological dispositions within a population and the extent to which such dispositions become institutionalized within political organizations may therefore be essential for understanding the practical consequences of authoritarianism.

Furthermore, although the multidimensional LWA measure demonstrated satisfactory psychometric properties, its content validity with respect to historical manifestations of LWA deserves further consideration. Many

of the items reflect contemporary ideological conflicts characteristic of liberal democracies, whereas several historical left-wing authoritarian movements emerged under markedly different social and political conditions. For example, the insurgency led by Sendero Luminoso in Peru was largely rooted in rural contexts and characterized by forms of revolutionary violence—including practices such as the *cuota de sangre* (i.e., blood quota), which required members to participate directly in acts of violence—that were considerably more extreme than the attitudes represented in the current instrument (Espinosa et al., 2017, 2023). Thus, while the scale appears to capture authoritarian tendencies expressed within contemporary progressive political discourse, it may not fully encompass the ideological and behavioral characteristics of historically influential left-wing authoritarian movements.

Even without extending this discussion to cross-cultural contexts, the reliance on a liberal framework also becomes apparent when the instrument is examined in light of the political science literature. For instance, Van Hiel et al. (2006) indicated that anarchists could not be regarded as left-wing authoritarians because, although they strongly oppose established authorities and existing power structures, they would not display elevated levels of authoritarian submission. Nevertheless, notable anarchist thinkers such as Bakunin, Kropotkin, Emma Goldman, and Errico Malatesta would also likely disagree with several items of the present instrument, particularly those comprising the Top-Down Censorship dimension, given their strong opposition to state suppression of opinions (Marshall, 2010). Likewise, considerable disagreement exists even within Marxist traditions, which according to this line of research could generally be classified as authoritarian. Most notably, Leon Trotsky harshly criticized the direction taken by Stalin's regime in the Soviet Union, arguing that the dictatorship of the proletariat should eventually be overcome through a process of political opening. At the same time, Trotsky himself defended the dictatorship of the proletariat during the October Revolution of 1917 (Deutscher, 1949). These examples illustrate that the categorization of authoritarianism within left-wing traditions is far from straightforward and suggest that the conception underlying the current LWA instrument is fundamentally rooted in a liberal understanding of authoritarianism. This limitation likely reflects the context in which the original instrument was developed. As noted by Altemeyer (1996), research on LWA in the United States has historically been shaped by a political context in which communist and revolutionary movements have exercised relatively little direct political influence. Consequently, the content of the instrument primarily reflects forms of LWA that are salient within contemporary political debates in the United States rather than those observed in historical revolutionary regimes or movements in other regions of the world.

It is important to note that the formulation of RWA demonstrates considerable conceptual consistency by linking its three components—Authoritarian Submission, Authoritarian Aggression, and Conventionalism—to relatively well-defined ideological content (Altemeyer, 1981). RWA has typically been associated with support for traditional family values, restrictive positions regarding sexual and gender freedoms, negative attitudes toward immigration, and, more broadly, the defense of a traditional moral order perceived as legitimate (Altemeyer, 1998). This does not imply that RWA encompasses the full ideological heterogeneity of the political right, nor that it necessarily represents an orientation aimed at maintaining the existing political system. Individuals high in RWA may, for example, adopt anti-establishment positions when they perceive that existing institutions have departed from the normative order they regard as legitimate (see for example Vilanova et al., 2027). Furthermore, the development of constructs such as SDO (Pratto et al., 1994) illustrates that other important dimensions of conservative political thought, particularly those concerning the acceptance of intergroup hierarchy and inequality, are not captured by RWA but remain theoretically relevant (Altemeyer, 1998). Thus, the relative conceptual consistency of RWA does not derive from an assumption that the political right is ideologically homogeneous, but rather from the relatively precise delineation of a particular configuration of authoritarianism and its associated ideological contents.

The subsequent development of specific measures of LWA can be understood, in part, as a reconsideration of Altemeyer's original position. Altemeyer (1996) questioned the need for a distinct measure of LWA, arguing that authoritarian dispositions could instead be identified using RWA among individuals on the political left. In this sense, proposals such as those of Conway et al. (2018) and Costello et al. (2022) represent an important contribution to the longstanding discussion regarding the existence and measurement of LWA. At the same

time, however, these operationalizations have been developed primarily within Western liberal democracies and emphasize manifestations of intolerance, censorship, and coercion associated with extreme liberal or progressive sectors. Consequently, they may capture only a subset of the historical and contextual manifestations that could potentially be understood as LWA.

Importantly, this issue cannot be resolved simply by defining LWA as opposition to, or support for, the existing political system. Different historical configurations of authoritarianism illustrate that authoritarian components may be expressed both in defense of an established system and in the pursuit of a radically different one. Stalinism in the USSR, for example, combined aggression toward perceived deviants, submission to authority, and strong ideological conventionalism within an established political system (Deutscher, 1949). In contrast, revolutionary movements such as Sendero Luminoso in Peru expressed comparable forms of aggression, authority, and ideological conventionalism from an explicitly anti-system position aimed at radically transforming the existing order (Comisión de la Verdad y Reconciliación, 2003). These contrasts suggest that a pro-system versus anti-system distinction does not, by itself, define the core of authoritarianism. Rather, conceptualizing LWA requires distinguishing between the structural components of authoritarianism and the specific ideological contents, authority referents, and conventions through which those components are expressed in particular historical and sociopolitical contexts.

Accordingly, one of the remaining challenges for the conceptualization and measurement of LWA is identifying left-wing ideological contents that allow authoritarian manifestations to be delineated with a degree of clarity comparable to that achieved for some contemporary manifestations of RWA. Whereas positions favoring traditional family values, restricting sexual and reproductive rights, or opposing immigration are relatively well-established ideological referents of contemporary RWA, it is less clear which specific ideological contents should consistently characterize LWA across societies and historical periods. This does not mean that LWA contents must be universal or constitute simple inversions of those associated with RWA. Rather, it requires distinguishing the authoritarian core (i.e., Authoritarian Submission, Authoritarian Aggression, and Conventionalism) from the concrete ideological referents toward which these components are directed. Because such referents may depend on the historical trajectory, power relations, and political traditions of a given society, measures of LWA require careful contextualization rather than assuming that manifestations identified in one political context necessarily represent LWA in another.

This conceptual issue is particularly relevant to the present measure. As discussed above, several items appear to capture contemporary ideological conflicts within liberal democracies rather than clearly delineated manifestations of the three components of authoritarianism. For example, some items assess attitudes toward political conservatives, while others concern censorship or social sanctions in specific institutional contexts. These features do not necessarily invalidate the measure, particularly given the psychometric evidence obtained in the present studies, but they raise questions regarding the scope of the construct being represented. More broadly, they suggest that future research on LWA should pay greater attention to the distinction between authoritarian processes and the ideological contents through which those processes are expressed.

It is also worth noting that the labels assigned to some of the LWA dimensions warrant further consideration. For instance, the Anti-Conventionalism dimension may not accurately reflect the content of the items it comprises. Rather than assessing opposition to conventional moral norms per se, several items explicitly target conservative individuals (e.g., “Conservatives are morally inferior to liberals”; “Deep down, just about all conservatives are racist, sexist, and homophobic”; “All political conservatives are fools”, Costello et al., 2022, p. 16). Consequently, the dimension appears to capture anti-conservative attitudes at least as much as, if not more than, opposition to conventionalism itself, thereby conflating ideological outgroup derogation with the rejection of traditional moral values. Future research should therefore seek to refine the construct by more clearly distinguishing between these two components and ensuring that all items target a common underlying dimension. A similar concern applies to the Top-Down Censorship dimension. Although its label implies support for censorship imposed by institutional authorities, not all items refer to top-down processes. Some instead assess support for bottom-up forms of social sanctioning and public pressure (e.g., “Colleges and universities that permit speakers

with intolerant views should be publicly condemned"; Costello et al., 2022, p. 16). Consequently, the current label may not adequately represent the full content of the dimension. Future refinements of the measure should therefore seek either to distinguish between institutional and grassroots forms of censorship or to adopt a broader label that more accurately reflects the construct being assessed.

Despite the aforementioned contributions, several limitations of our studies should be acknowledged. First, although the present studies included participants spanning different points of the political spectrum, the psychometric properties of the abbreviated LWA measure were not specifically examined among individuals who self-identify with organizations occupying the far-left of the Brazilian political landscape as Van Hiel et al. (2006) did in the Flemish context. Future studies could investigate whether the factor structure, reliability, and validity of the instrument are replicated among individuals affiliated with parties and movements explicitly endorsing more radical left-wing positions. Second, although the Top-Down Censorship dimension demonstrated acceptable internal consistency across Studies 1 and 2, its reliability coefficients were somewhat lower than those observed for the remaining dimensions. One possible explanation is that support for restricting certain forms of expression may reflect concerns that extend beyond left-wing authoritarian tendencies and may be endorsed by individuals across different ideological positions. However, this possibility remains speculative and should be examined directly in future research. Third, the use of convenience sampling through social media in both studies may limit the representativeness of the samples and, consequently, the generalizability of the findings to the broader adult and adolescent populations. Fourth, the short LWA measure was intentionally designed to include only three indicators per dimension. Although this decision necessarily reduced the breadth of item coverage within each construct, a minimum of three indicators per latent factor is generally recommended to ensure adequate identification and stability of factor solutions (Bollen, 1989; Raubenheimer, 2004). The three items for each dimension were selected based on their factor loadings in the ESEM conducted in Study 1; consequently, the factor loadings and model fit observed for the resulting 9-item measure in this same sample may have been favourably influenced by this item-selection procedure and should therefore be interpreted with some caution. Importantly, however, the three-factor structure of the resulting measure was subsequently evaluated in the independent adolescent sample in Study 2 through CFA, providing a separate assessment of the proposed structure. Moreover, all standardized factor loadings exceeded 0.60 in both studies, a threshold typically regarded as indicative of acceptable practical significance in psychometric research (Hair et al., 2019). Taken together with the satisfactory fit of the three-factor CFA model in Study 2, these findings suggest that the brevity of the instrument did not compromise its psychometric adequacy, while acknowledging that the initial item selection and validation were conducted in the same adult sample. It is important to note that the three-factor CFA model was overidentified ($\chi^2 = 21.398$, $df = 24$), allowing for the empirical assessment of model fit. The CFI and TLI values of 1.00 and the RMSEA of 0.001 reflect the close correspondence between the hypothesized model and the observed data. In particular, the chi-square statistic was lower than the degrees of freedom, resulting in an RMSEA estimate close to zero. However, these values should not be interpreted as indicating the complete absence of measurement error. The three-factor structure was specified a priori, based on the theoretical structure of the LWA instrument, and modification indices were not used to guide model specification. Consequently, no residual covariances or cross-loadings were added on the basis of modification indices. Importantly, the replication of the three-factor structure across adult and adolescent samples should not be interpreted as evidence of factorial invariance or measurement equivalence between these populations, which should be formally examined in future studies administering the same version of the instrument to both age groups. At last, 23 participants with 80–99% of survey items completed had missing data imputed using the series-mean method; although this represented a small proportion of the sample, this approach may have introduced some bias into the analyses.

Conclusion

To the best of our knowledge, this study provides the first adaptation and validation of a measure of LWA in Brazilian Portuguese. The findings indicate that the proposed 9-item version preserves the multidimensional nature of the original instrument while demonstrating satisfactory psychometric properties among both adults and adolescents. By substantially reducing administration time without compromising reliability and validity, the abbreviated scale offers researchers a practical alternative for the assessment of LWA in large-scale surveys, studies including multiple constructs, and investigations involving younger populations. More broadly, the availability of a brief and psychometrically sound measure may facilitate the incorporation of LWA into the Brazilian literature on political attitudes and ideological processes, thereby contributing to a more balanced and empirically grounded understanding of authoritarian tendencies across the political spectrum. The instrument is available at (https://www.osf.io/7p3xj/overview? view_only=a96f93fea88b4e08a88bbb99914e3b58).

Abbreviations

AC	Anti-Conventionalism
ACT	Authoritarianism-Conservatism-Traditionalism
AHA	Anti-Hierarchical Aggression
C.I.	Confidence Interval
CFA	Confirmatory Factor Analysis
CFI	Comparative Fit Index
Df	Degrees of Freedom
ESEM	Exploratory Structural Equation Model
LWA	Left-Wing Authoritarianism
M	Mean
NEP	New Environmental Paradigm
RMSEA	Root Mean Square Error of Approximation
RWA	Right-Wing Authoritarianism
SD	Standard Deviation
SDO	Social Dominance Orientation
TDC	Top-Down Censorship
TLI	Tucker-Lewis Index
WLSMV	Weighted Least Squares Mean and Variance adjusted
SRMR	Standardized Root Mean Square Residual
WRMR	Weighted Root Mean Square Residual

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Authors' contributions FV played a lead role in conceptualization, software, validation, formal analysis, data curation, visualization, and writing of original draft and an equal role in methodology, investigation, and writing of review and editing. TLM played an equal role in conceptualization, writing of original draft, and writing of review and editing and a supporting role in methodology, investigation, funding acquisition, and supervision. AE played an equal role in writing of original draft, and writing of review and editing, and a supporting role in methodology. ABC played a lead role in resources, supervision, project administration, and funding acquisition and an equal role in conceptualization, methodology, investigation, and writing of original draft. All authors read and approved the final manuscript.

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Data availability Datasets of Studies 1 and 2, as well as the final version of the instrument are available at (https://osf.io/7p3xj/overview?view_only=a96f93fea88b4e08a88bbb99914e3b58).

Declarations

Ethics approval and consent to participate The design of Study 1 was approved by the ethics committee of the Pontifícia Universidade Católica do Rio Grande do Sul under registration number CAAE 51664121.8.0000.5336 in accordance with the Resolution 510/2016 of the Brazilian National Health Council. Before answering the instruments, participants expressed their consent by providing their agreement to an Informed Consent Form. Anonymity was granted and only researchers had access to the data. The design of Study 2 was also approved by the ethics committee of the Pontifícia Universidade Católica do Rio Grande do Sul under registration number CAAE 51664121.8.0000.5336 in accordance with the Resolution 510/2016 of the Brazilian National Health Council. All data collection procedures, including the fact that the written informed consent of participants' legal guardian was waived, are according to the Institutional Review Board procedures approved under the aforementioned registration number and the Resolution 510/2016 of the Brazilian National Health Council. Also in Study 2, before answering the instruments, participants expressed their consent by providing their agreement to an Informed Consent Form. Anonymity was granted and only researchers had access to the data.

Competing interests The authors declare no competing interests.

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